

3rd Grade – Noetic-style Practice Test (Spring)

Time: 45 minutes • Answer all questions. Do not use a calculator.

1. Sofia had 248 stickers. She gave 37 stickers to a friend and then put the rest into packets of 6 stickers each. How many full packets did she make?
2. A cake recipe calls for $\frac{3}{4}$ cup sugar to make 8 cupcakes. How much sugar is needed per cupcake? How many cupcakes can you make with $2\frac{1}{4}$ cups of sugar?
3. A bus picks up 18 students at the first stop, 23 at the second stop, and twice as many as the total from the first two stops at the third stop. How many students are on the bus after the third stop?
4. Write the largest 3-digit number that is a multiple of 7. Then write the largest 3-digit number that is a multiple of 9.
5. Liam has coins that are only quarters (25¢) and dimes (10¢). He has 11 coins totaling \$2.15. How many quarters and how many dimes does he have?
6. Draw a rectangle with integer side lengths whose area is 36. Give one possible pair of side lengths.
7. There are 5 red, 4 blue, and 3 green pens in a box. If you pick one pen without looking, what is the probability it is not blue? Give your answer as a fraction in lowest terms.
8. A train leaves at 9:15 AM and arrives at 12:05 PM. How many minutes long is the trip?
9. A library has 24 books and 6 shelves. It places the books equally on each shelf. After adding 6 more books, it still places books equally on the 6 shelves. How many books were on each shelf before and after adding the 6 books?
10. If $3 \times n = 81$, what is n ? Show how you know. Then find $81 \div 3$ using mental math and confirm it matches.
11. A garden bed is 5 meters long and 3 meters wide. A path 1 meter wide runs all the way around inside the garden (parallel to the sides). What is the area of the path?

12. Write all the two-digit numbers you can make using the digits 1, 2, and 3 without repeating digits. How many are even? What is their sum?

13. A puzzle: I think of a number. I subtract 9, then divide the result by 4, and get 7. What was the original number?

14. A pattern of shapes goes: $\blacktriangle \blacktriangle \blacksquare \blacktriangle \blacktriangle \blacksquare \blacktriangle \blacktriangle \blacksquare \dots$ What is the 25th shape in the pattern?

15. Three friends share \\$9.00 in the ratio 2 : 3 : 4. How much money does each friend receive? (Give answers in dollars and cents.)

16. A rectangular poster is 40 cm by 30 cm. If you place square stickers of side 5 cm (no overlap) to cover as much of the poster as possible, how many stickers fit entirely on the poster?

17. A farmer packs apples into boxes of 8. He fills 14 boxes and has 5 apples left over. How many apples did he start with?

18. On a clock, the minute hand points at 4. What time is it? (Give the time in minutes past the hour.)

19. Maya reads 12 pages on Monday, 15 pages on Tuesday, and 18 pages on Wednesday. If she increases the number of pages by 3 each day, how many pages does she read on Sunday (the 7th day) if Monday is day 1?

20. A rectangle is divided into a grid of 3 rows and 5 columns of equal small rectangles. One small rectangle is shaded. What fraction of the whole is shaded? If the big rectangle has perimeter 32 cm and the longer side is twice the shorter side, what is the area of the shaded small rectangle?